Qwen2.5-VL-3B: Zero-Shot Robust vs Blended Prompting on CIFAR-10 Corruptions

VizInt Experiment

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Abstract

We evaluate Qwen/Qwen2.5-VL-3B-Instruct (4-bit quantized) as a zero-shot classifier on CIFAR-10 with image corruptions. Two prompting strategies are compared: **Robust** (classify only) and **Blended** (identify corruption then classify). Evaluated on 1000 images per condition.

1 Method

The Robust prompt asks the VLM to classify into CIFAR-10 categories via JSON output. The Blended prompt additionally asks the model to identify the corruption type (none, gaussian_noise, rotation, affine) before classifying, mirroring how auxiliary transform prediction heads improve traditional models.

2 Results

Table 1: Classification accuracy at various corruption severities.

Model	Clean	S=0.25	S=0.5	S=0.75	S=1.0	Avg Drop	OOD
Qwen2.5-VL-3B Robust	0.8270	0.5860	0.5090	0.4690	0.4660	0.3195	0.3890
Qwen2.5-VL-3B Blended	0.7810	0.4970	0.4480	0.4400	0.4130	0.3315	0.3390

Table 2: Per-transform accuracy at S=1.0.

Model	Gaussian Noise	Rotation	Affine
Qwen2.5-VL-3B Robust	0.1119	0.2996	0.6506
Qwen2.5-VL-3B Blended	0.0702	0.2622	0.5510

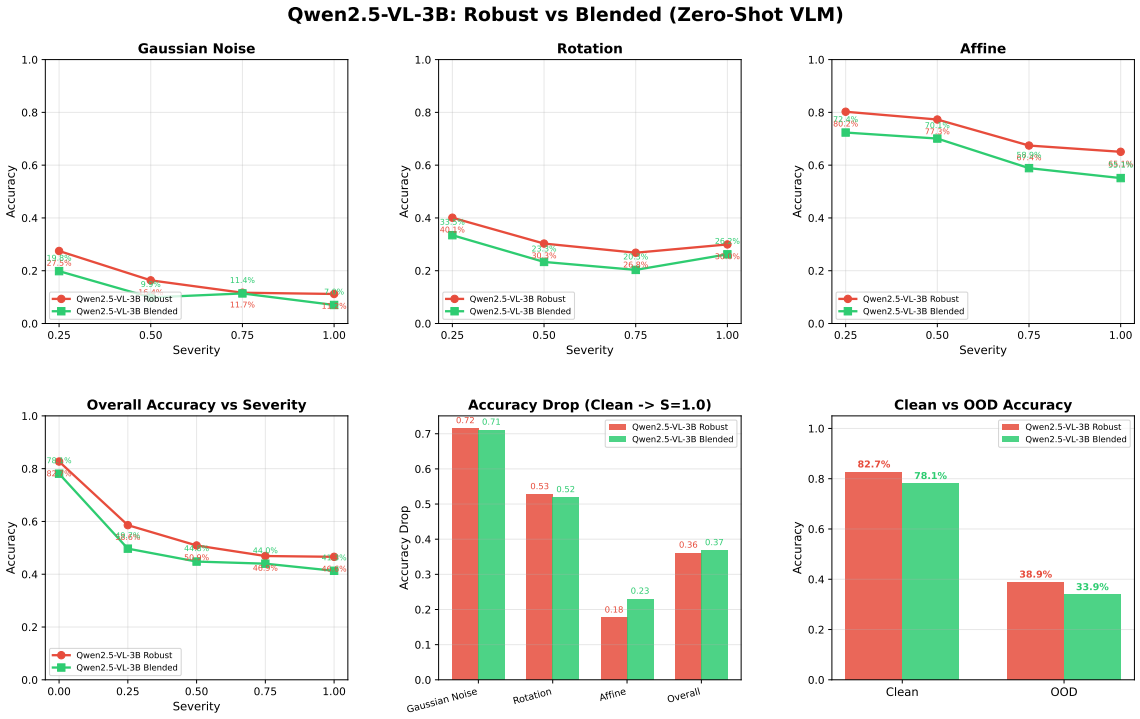


Figure 1: Qwen2.5-VL-3B: Robust vs Blended zero-shot prompting comparison.